

Jason D. Lee

Curriculum Vitae

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RESEARCH INTERESTS

Statistics: Post-model selection inference, High-dimensional estimation and model selection

Machine Learning: Large-scale matrix factorizations, Distributed and communication-efficient sparse learning

Large-scale Optimization: Distributed optimization, Non-smooth optimization, Stochastic optimization

EDUCATION

2010–present **Stanford University, Computational and Mathematical Engineering.**

Ph.D. Candidate

- Adviser: Trevor Hastie
- Co-Adviser: Jonathan Taylor
- GPA: 4.13

2006–2010 **Duke University, Department of Mathematics.**

Bachelor of Science in Mathematics

- Graduation with High Distinction, Phi Beta Kappa, Magna Cum Laude*
- Adviser: Mauro Maggioni

2002–2006 **Lynbrook High School, San Jose, CA.**

RESEARCH EXPERIENCE

2011–Current **Research with Trevor Hastie, Stanford University.**

2011–Current **Research with Jonathan Taylor, Stanford University.**

Summer 2012 **Intern with Ran Gilad-Bachrach, Microsoft Research Redmond.**

Summer 2011 **Intern with Emre Kiciman, Microsoft Research Redmond.**

Summer 2010 **Intern with Nati Srebro, Toyota Technology Institute Chicago.**

Summer 2009 **Research with Mauro Maggioni, Duke University.**

PUBLICATIONS

JOURNAL PUBLICATIONS

Jason D. Lee, Yuekai Sun, and Michael Saunders. Proximal Newton-type Methods for Minimizing Convex Objective Functions in Composite Form. *SIAM Journal on Optimization*, 24(3): 1420-1443, 2014.

Jason D. Lee and Trevor Hastie. Learning the structure of mixed graphical models. *Journal of Computational and Graphical Statistics*, 2014

Jason D. Lee, Dennis Sun, Yuekai Sun, and Jonathan Taylor. Exact post-selection inference with the lasso. *arXiv:1311.6238*, *Under revision*.

Jason D. Lee, Yuekai Sun, and Jonathan Taylor. Model selection consistency of M-estimators with geometrically decomposable penalties. *Preliminary version at NIPS 2013*, *arXiv:1305.7477*, *Under revision*.

Jason D. Lee, Yuekai Sun, and Jonathan Taylor. Evaluating the statistical significance of biclusters. *Submitted*.

Trevor Hastie, Rahul Mazumder, **Jason D. Lee**, and Reza Zadeh. Softimpute-ALS: a fast alternating least squares algorithm for distributed large-scale matrix completion. *Submitted*.

REFEREED CONFERENCE PUBLICATIONS.

Austin R. Benson, **Jason D. Lee**, Bartek Rajwa, and David F. Gleich. Scalable methods for nonnegative matrix factorizations of near-separable tall-and-skinny matrices. *Neural Information Processing Systems (NIPS)*, 2014. Spotlight presentation.

Jason D. Lee and Jonathan Taylor. Exact statistical inference for marginal screening. *Neural Information Processing Systems (NIPS)*, 2014.

Jason D. Lee, Ran Gilad-Bachrach, and Rich Caruana. Using multiple samples to learn mixture models. *Neural Information Processing Systems (NIPS)*, 2013. Spotlight presentation.

Jason D. Lee, Yuekai Sun, and Jonathan Taylor. On model selection consistency of M-estimators with geometrically decomposable penalties. *Neural Information Processing Systems (NIPS)*, 2013.

Jason D. Lee and Trevor Hastie. Learning Mixed Graphical Models. *AI and Statistics (AISTATS)*, 2013.

Jason D. Lee, Yuekai Sun and Michael Saunders. Proximal Newton-type Methods for Minimizing Convex Objective Functions in Composite Form. *Neural Information Processing Systems (NIPS)*, 2012.

Jason D. Lee, Benjamin Recht, Ruslan Salakhutdinov, Nathan Srebro and Joel Tropp. Practical Large-Scale Optimization for Max-Norm Regularization. *Neural Information Processing Systems (NIPS)*, 2010.

Jason D. Lee and Mauro Maggioni. Multiscale Dynamic Graphs. *Sampling Theory and its Applications*, 2011.

Markus Kliegl, **Jason D. Lee**, Jun Li, Xinchao Zhang, David Rincon, and Chuanxiong Guo. Generalized DCell Network Structure for Load-Balanced Data Center Networks. *IEEE Infocom 2010*.

Anna V. Little, **Jason D. Lee**, Yoon-Mo Jung, and Mauro Maggioni. Estimation of Intrinsic Dimensionality of Samples from Noisy Low-Dimensional Manifolds in High Dimensions with Multiscale Svd. *IEEE Statistical Signal Processing 2009 (SSP 2009)*.

WORKING PAPERS

Jason D. Lee, Qiang Liu, Yuekai Sun, and Jonathan Taylor. One shot distributed sparse regression and regularized M-estimation. *In Preparation*.

Austin Benson, **Jason D. Lee** and Yuekai Sun. Distributed proximal Newton methods for sparse regression and M-estimation: Faster convergence with more data. *In Preparation*.

TECHNICAL REPORTS

Jason D. Lee, Yuekai Sun and Michael Saunders. Inexact Analysis of Proximal Newton-type Methods. *Neural Information Processing Systems (NIPS) 2012, Workshop on Optimization and Machine Learning*.

Jason D. Lee and John Neuberger. Existence of Asymptotic Solutions to Semilinear Partial Difference Equations on Graphs. <http://jan.ucc.nau.edu/~ns46/student/lee.reu.2007.pdf>, *AMS-MAA Joint Mathematics Meetings 2008*.

Jason D. Lee. Estimation of Intrinsic Dimensionality of Point Cloud Data and Multiscale Analysis of Dynamic Graphs. *Senior Thesis*.

Markus Kliegl, **Jason D. Lee**, Jun Li, Xinchao Zhang, David Rincon, and Chuanxiong Guo. The Generalized DCell Network Structures and Their Graph Properties. <http://research.microsoft.com/apps/pubs/default.aspx?id=103129>, *Microsoft Research Technical Report*.

SOFTWARE

PNOPT, *Package for Proximal Newton-type methods for non-smooth optimization*, Jason D. Lee, Michael Saunders, and Yuekai Sun, <https://github.com/yuekai/PNOpt>.

Nonnegative matrix factorizations in MapReduce, *Implementation of Scalable methods for nonnegative matrix factorizations of near-separable tall-and-skinny matrices*, Austin R. Benson, Jason D. Lee, Bartek Rajwa, and David F. Gleich, <https://github.com/arbenson/mrnmf>.

Gradient and Newton methods for structure learning of mixed graphical models, *Matlab implementation for structure learning of mixed graphical models*, Jason D. Lee and Trevor Hastie, <http://web.stanford.edu/~jdl17/learningmgm.html>.

TEACHING EXPERIENCE

Winter 2013 **Instructor**, *Introduction to Scientific Computing with MATLAB (CME 192)*, Stanford University.

- Winter 2013 **Organizer**, *Data Science Colloquium Series, Department of Statistics and Institute of Computational and Mathematical Engineering*, Stanford University.
- Spring 2009 **Teaching Assistant**, *Combinatorics*, Duke University.

AWARDS AND HONORS

- 2010 **NSF Graduate Fellowship in Applied Mathematics**.
Awarded to 1500 PhD students across all science, engineering and humanities.
- 2010 **National Defense Science and Engineering Graduate Fellowship (NDSEG)**.
Awarded to 200 PhD students across all fields of science and engineering.
- 2010 **Phi Beta Kappa**, *Duke University Chapter*.
- 2010 **Stanford Graduate Fellowship, Office of Technology Licensing Fellow**, *Stanford University*.
Awarded to 100 PhD students at Stanford University.
- 2010 **Awarded Graduation with High Distinction**, *Duke University*.
Awarded to top 2 graduates of Duke Mathematics.
- 2010 **Duke Dean's List/Dean's List with Distinction for 8 consecutive semesters**.
Awarded for GPA in the top 10% of class.
- 2010 **Julia Dale Prize in Mathematics**, *Duke University*.
Awarded to top graduate of Duke Mathematics Department

SELECTED TALKS

- 08/2014 **Exact inference after model selection**, *Joint Statistics Meetings 2014*.
- 08/2014 **Exact inference after model selection with lasso**, *Information Initiative at Duke Seminar*, Duke University.
- 06/2014 **Learning mixed graphical models**, Workshop on covariance selection, International Conference on Machine Learning 2014.
- 05/2014 **Large-scale proximal Newton methods for composite optimization**, *Alpine Data Labs*.
- 01/2014 **Learning mixed graphical models**, *Modern Large Scale Statistical Learning Workshop*.
- 3/2013 **Large-scale proximal Newton methods for composite optimization**, *Management Sciences Seminar*, Stanford.
- 09/2011 **Distributed Algorithms for Matrix Factorization and Singular Value Decomposition**, *Microsoft Research Redmond*.

REVIEWING

Reviewer for ICML, AISTATS, and NIPS conferences (40 papers).

Reviewer for Journal of Machine Learning Research, AOAS, Biometrika, and Math Programming.